Single-LLM Versus Multi-Agent Debate Systems for Scientific Hypothesis Generation

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Large language models (LLMs) are increasingly being used to propose scientific hypotheses, retrieve and synthesize literature, critique candidate explanations, and support experimental design. Two broad paradigms have emerged: single-LLM systems that rely on one model with prompting, retrieval, self-reflection, or tool use; and multi-agent systems in which several model instances interact through debate, role specialization, iterative review, or consensus formation. This review compares these paradigms for scientific hypothesis generation. Rather than treating multi-agent interaction as automatically superior, the literature is organized around five themes: hypothesis generation and novelty, evidence grounding and verification, critique and debate, evaluation of scientific quality, and integration with autonomous research workflows. The reviewed studies suggest that single-LLM systems offer simplicity, lower cost, easier control, and strong performance when supplied with suitable literature or domain tools. Multi-agent systems can improve exploration, criticism, and the explicit comparison of competing ideas, particularly when agents are genuinely diverse and are assigned complementary roles. However, debate introduces substantial costs and can amplify shared model errors, sycophancy, premature consensus, and problem drift. The strongest direction is therefore not a binary choice but a conditional hybrid architecture: independent generation followed by evidence retrieval, targeted adversarial review, selective debate, and experimental validation. Important gaps remain in standardized benchmarks, causal attribution of multi-agent gains, calibration, diversity measurement, and real-world laboratory validation.

Additional Key Words and Phrases: large language models, scientific hypothesis generation, multi-agent systems, debate, literature-based discovery, autonomous science, evaluation

1 Introduction

Scientific discovery depends on the ability to connect observations, identify regularities, formulate explanatory hypotheses, and determine which hypotheses deserve empirical testing. Computational approaches to discovery have a long history. Early systems such as BACON, a computational scientific-discovery system designed to identify regularities in observed data and formulate candidate scientific laws, demonstrated that programs could search observations for regularities and express them as candidate scientific laws, while literature-based discovery showed that useful knowledge can emerge by connecting information that is distributed across otherwise separate bodies of research [12, 20].

Recent LLMs have changed the scale and flexibility of this problem. Instead of relying only on manually designed discovery rules, an LLM can summarize literature, propose mechanisms, combine concepts, generate experiments, and revise ideas in natural language. Studies have examined both direct hypothesis generation [24, 25] and broader research automation, including literature retrieval, coding, experimentation, and manuscript preparation [13, 23]. A single-LLM approach can also provide a relatively simple and lower-cost way of supporting scientific reasoning, functioning as an accessible assistant that can generate and refine ideas without requiring the coordination of multiple agents.

At the same time, researchers have begun to replace the single model with groups of interacting agents. In these systems, one agent may generate ideas, another may act as a critic, and others

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may assess novelty, feasibility, evidence, or experimental design. Multi-agent debate is attractive because scientific reasoning is not only a process of generating explanations; it is also a process of challenging them. Multiple agents can introduce different perspectives and provide repeated opportunities for critique and refinement. However, this interaction increases computational and engineering costs and may also produce agreement without correctness, particularly when agents share similar biases or reinforce unsupported claims.

These trade-offs suggest that the comparison need not be limited to choosing between a single LLM and a fully multi-agent system. The lower cost and simplicity of single-LLM approaches and the diversity and critical evaluation offered by multi-agent systems could potentially be combined or selectively optimized. A hybrid approach could therefore use the appropriate level of interaction depending on the scientific task, research question, or stage of hypothesis generation. The central question is therefore not simply whether multiple agents outperform one, but under what conditions different combinations of single-LLM and multi-agent capabilities are most suitable for scientific hypothesis generation.

This review addresses that question through the following research question: **How do single-LLM and multi-agent debate systems differ in their ability to generate novel, valid, feasible, and experimentally useful scientific hypotheses, and what design choices determine their relative advantages?** The review focuses on systems that produce scientific ideas or hypotheses, while also considering adjacent work on reasoning, self-reflection, scientific agents, and automated experimentation.

1.1 Scientific hypothesis generation with LLMs

A scientific hypothesis can be viewed as a structured, testable proposition that links observations, variables, mechanisms, or interventions. In practical research, a useful hypothesis must satisfy several conditions at once: it should be sufficiently novel, connected to existing evidence, scientifically plausible, operationally testable, and potentially consequential. These criteria can conflict. An idea that is highly novel may be weakly grounded; an idea that is strongly supported by existing literature may be incremental.

Computational discovery systems traditionally separated the search for candidate regularities from the evaluation of those candidates. BACON-like systems searched data for patterns and formulated laws [12]. Literature-based discovery approaches instead searched textual or bibliographic spaces for meaningful connections [20]. LLM-based systems combine aspects of these functions through a flexible language interface, but their hypothesis-generation mechanism is fundamentally different from conventional rule-based discovery.

An LLM generates text by estimating a probability distribution over possible next tokens conditioned on the preceding context. In simplified form, the probability of a generated sequence can be expressed as

$$P(y_{1:T} \mid x) = \prod_{t=1}^{T} P(y_t \mid y_{<t}, x), \tag{1}$$

where x denotes the input context, y_t is the token generated at position t, and $y_{<t}$ represents the preceding generated tokens. Consequently, hypothesis generation does not require the model to explicitly discover a scientific law or perform a formally valid inference. Instead, the model can generate candidate explanations by recombining patterns and relationships represented in its learned parameters and the supplied context.

This raises an important question: is probabilistic text generation scientifically sound as a basis for hypothesis generation? The answer depends on what is meant by "sound." Probabilistic generation by itself does not establish that a hypothesis is scientifically true, causally valid, or even factually

correct. A highly probable continuation may simply reflect patterns frequently present in the training data. LLMs can therefore produce hypotheses that are plausible in language while being unsupported by evidence. This limitation is particularly important because the probability assigned to a generated sequence should not be interpreted as a probability that the scientific hypothesis itself is true.

However, scientific hypothesis generation is inherently exploratory. A hypothesis is a proposition to be tested rather than a conclusion that is already known to be correct. Recent work therefore argues that the probabilistic and sometimes unexpected outputs of LLMs can be useful for proposing candidate hypotheses, provided that those candidates are subsequently evaluated using scientific evidence or experiments. Zhou et al. demonstrate that LLM-generated hypotheses can be iteratively updated from observed data and can improve predictive performance over simple few-shot prompting in several benchmark settings [25]. Laboratory validation provides stronger evidence for the feasibility of this approach: GPT-4 generated candidate drug combinations, several of which subsequently showed positive experimental synergy, although the study also reported limitations in the quality of the explanations accompanying the generated hypotheses [4].

Thus, the scientific value of an LLM-generated hypothesis should not be judged from the generation probability alone. The LLM can be viewed as a mechanism for exploring a large space of candidate ideas, while novelty, evidential grounding, plausibility, feasibility, and experimental validity must be assessed separately. This distinction is central to comparing single-LLM systems with multi-agent approaches: the main question is not which architecture generates the most plausible text, but which architecture more reliably converts generated candidates into scientifically useful hypotheses.

1.2 Single-LLM systems

A single-LLM system uses one model instance as the principal generator and reasoner. It may be enhanced through prompt engineering, chain-of-thought-style reasoning, retrieval-augmented generation, external tools, memory, self-critique, or iterative refinement. Self-Refine and Reflexion illustrate the broader idea that a single model can generate an output, inspect it, and revise it using feedback or memory [14, 16].

The single-LLM paradigm should therefore not be equated with a single prompt. A single model can perform multiple stages internally or sequentially. The defining feature is that the stages are controlled by one model rather than by independently instantiated agents with potentially different roles or instructions.

1.3 Multi-agent debate systems

Multi-agent systems distribute the process across several model instances or agent roles. A typical workflow includes independent proposal, critique, rebuttal, synthesis, and ranking. Debate may be open-ended, role-based, or mediated by a coordinator. The motivation is to increase search breadth and create explicit disagreement before a final decision.

Multi-agent debate was initially studied for factuality and reasoning rather than scientific ideation specifically. Du et al. reported that debate can improve performance on some reasoning and factuality tasks [7]. However, later analysis showed that multi-agent debate does not reliably dominate simpler baselines and is sensitive to configuration, cost, and task characteristics [18]. These findings are directly relevant to scientific hypothesis generation, where the quality of the evaluation process may matter more than the number of agents.

2 Thematic Literature Review

2.1 Theme 1: Generation, novelty, and exploration

One major contribution of LLMs is their ability to generate candidate research directions from large amounts of scientific language. Zhou et al. study hypothesis generation with LLMs and show how prompting and data-driven formulations can be used to produce candidate hypotheses [25]. Yang et al. similarly investigate automated open-domain scientific hypothesis discovery, demonstrating the potential of LLMs to move beyond narrow question answering toward research-oriented generation [24].

However, novelty is difficult to define and measure. A generated statement may be linguistically unusual while being scientifically trivial, already known, or impossible to test. SciMON (Scientific Inspiration Machines for Optimizing Novelty) addresses this issue by retrieving scientific inspirations and optimizing candidate ideas for novelty [22]. Its design implies that novelty should be treated as a search objective informed by literature, not as a property inferred only from the wording of the output.

Human evaluation studies reinforce this distinction. Si et al. evaluate LLM-generated research ideas with a large group of NLP researchers and examine dimensions such as novelty and feasibility [17]. Their work suggests that LLMs can produce ideas that are competitive in some respects, but it also shows why expert assessment remains necessary: research quality involves implicit disciplinary knowledge and judgments about whether an idea would actually advance the field.

Multi-agent systems potentially improve exploration by allowing several agents to propose alternatives before synthesis. Su et al. report that a multi-agent scientific-ideation system can improve idea quality relative to single-agent baselines under their evaluation protocol [19]. The likely mechanism is not merely parallel text generation. Different agents can search different conceptual paths, expose assumptions, and preserve alternatives that a single generation might discard too early.

Nevertheless, more agents do not automatically imply more useful diversity. If all agents share the same model, prompts, retrieval corpus, and priors, their proposals may be highly correlated. Diversity must therefore be designed through independent sampling, role variation, different evidence views, or explicit disagreement objectives.

2.2 Theme 2: Evidence grounding, literature retrieval, and verification

Scientific hypotheses cannot be evaluated solely by fluency. They must be connected to observations, prior findings, mechanisms, and feasible measurements. Literature grounding is consequently central to both single- and multi-agent systems. Zhou et al. emphasize the importance of using scientific data and context in hypothesis generation [25], while SciMON explicitly retrieves literature-based inspirations to guide novelty [22].

Evidence grounding can improve factuality and reduce the risk that an LLM invents unsupported mechanisms. It also makes the generated hypothesis more useful to researchers because the system can identify supporting findings, conflicting evidence, and missing experiments. However, retrieval alone does not guarantee correctness. A model may misinterpret a paper, cite evidence that does not entail the claim, or combine results from incompatible settings.

The distinction between single-LLM and multi-agent approaches is therefore not necessarily a distinction between what a model can and cannot do. A single LLM can be organized into sequential stages in which it retrieves evidence, generates a hypothesis, critiques the proposal, and revises it. Multi-agent architectures make these roles explicit by assigning them to separate interacting agents. Such separation can provide specialized prompts, independent critiques, different contexts, or parallel reasoning paths, but it also introduces additional computational and coordination costs.

The potential advantage of multi-agent systems therefore depends on whether these additional sources of diversity and specialization improve the evaluation process beyond what can be achieved through sequential use of a single model.

The AI co-scientist line of work illustrates a broader generate–debate–evolve pattern, where candidate ideas are repeatedly assessed and refined [9]. The Virtual Lab similarly demonstrates the value of connecting agentic reasoning to experimental validation rather than treating textual agreement as the final endpoint [1]. These approaches do not necessarily imply that multiple agents are always required; rather, they illustrate how structured interaction and iterative evaluation can be incorporated into scientific discovery.

Both approaches can also inherit the same fundamental evidence-related limitations. If the initial retrieval is incomplete, a single-LLM pipeline or every downstream agent may reason over the same missing evidence. Likewise, additional agents may multiply rather than correct errors if they rely on the same unsupported information. A robust architecture should therefore record provenance, distinguish retrieved evidence from model inference, and require explicit statements of uncertainty.

2.3 Theme 3: Critique, self-reflection, and debate

Critique is the strongest conceptual argument for multi-agent debate. Scientific reasoning requires identifying hidden assumptions, alternative explanations, confounding variables, and weaknesses in experimental design. Single-LLM systems can perform self-critique through iterative prompting. Self-Refine uses feedback-driven revision [14], and Reflexion adds verbal feedback and memory to support future attempts [16]. These methods can be effective when the model is capable of recognizing its own errors.

However, self-critique is not independent criticism. The same model that generated a hypothesis also supplies the criteria, interpretation, and final judgment. It may rationalize its first answer instead of genuinely challenging it. Multi-agent debate addresses this problem by separating proposal and criticism across agents. Du et al. provide general evidence that debate can improve some reasoning and factuality outcomes [7].

Scientific ideation introduces additional complexity. Debate participants may disagree about novelty, feasibility, or importance even when they agree on the underlying facts. Smit et al. caution that multi-agent debate is not universally better than self-consistency or ensembling and that performance depends on the debate strategy and task [18]. This is especially important for scientific hypotheses because there is often no immediate ground-truth answer.

Debate can also create undesirable social dynamics among agents. Agents may converge on an early proposal because it is repeated frequently, appears authoritative, or is presented with high confidence. Consensus can therefore be a poor proxy for scientific validity. The most useful debate design is likely adversarial but structured: agents should be required to state evidence, identify falsification conditions, propose alternatives, and preserve unresolved disagreement.

2.4 Theme 4: Evaluation of scientific quality

Evaluation remains the central bottleneck in the literature. Common dimensions include novelty, relevance, significance, feasibility, validity, clarity, and potential impact. These dimensions are related but not interchangeable. A hypothesis can be novel but implausible, feasible but unimportant, or significant but already established.

IdeaBench proposes a benchmark-oriented approach to research-idea generation and emphasizes systematic evaluation of research ideas [10]. The biomedical evaluation by Qi et al. similarly considers multiple dimensions, including novelty, relevance, significance, and verifiability [15]. Such work is important because it moves evaluation away from a single LLM-judge score.

Experimental validation provides a stronger form of evidence. Boiko et al. demonstrate autonomous chemical research with LLMs, showing how language-based reasoning can be connected to laboratory procedures [5]. The breast-cancer laboratory study provides another example of LLM-generated drug-combination hypotheses being tested experimentally [3]. These studies do not establish that one architecture is universally superior, but they show that scientific usefulness ultimately depends on whether generated ideas survive contact with experiments.

Multi-agent systems may improve evaluation by assigning separate agents to novelty, feasibility, evidence, and risk. However, this decomposition can create false precision. Scores from several agents may look rigorous while still reflecting the same model biases. Evaluation should therefore include external evidence, expert judgment, prospective prediction, and where possible, experimental success. These evaluation challenges become particularly important as hypothesis generation is incorporated into broader autonomous research workflows.

2.5 Theme 5: From hypothesis generation to autonomous science

The literature increasingly treats hypothesis generation as one stage in an end-to-end research loop. The AI Scientist attempts to automate idea generation, coding, experimentation, analysis, and paper writing [13]. The AI co-scientist and related systems extend this idea through multiple agents, iterative debate, and evidence-based evolution [9].

These systems suggest that the single-versus-multi distinction is becoming less clear in practice. A single agent may call tools and perform multiple internal roles, while a multi-agent system may use a coordinator that effectively behaves like a single decision-maker. The more useful comparison is therefore functional: how many independent searches occur, how criticism is generated, how evidence is checked, and how final decisions are validated.

End-to-end automation also exposes limitations. Autonomous systems may generate underdeveloped ideas, implement experiments incorrectly, or produce manuscripts that appear coherent despite methodological weaknesses [13, 23]. This reinforces the need for human oversight and explicit validation gates.

3 Comparison of Existing Approaches

Single-LLM and multi-agent systems should be compared across several dimensions rather than by one overall score. Table 1 summarizes the main differences identified in the literature across cost, search breadth, criticism, evidence use, reproducibility, scalability, and validation.

- **Cost and efficiency.** Single-LLM systems generally require fewer model calls and are easier to deploy. Multi-agent debate increases token use, latency, orchestration complexity, and often the number of retrieval operations.
- **Search breadth.** A single model can generate multiple samples, but multi-agent systems make parallel exploration explicit. Their advantage depends on whether agents are genuinely diverse.
- **Criticism.** Self-reflection is cheap and easy to integrate, but multi-agent criticism can provide more independent perspectives. Neither guarantees correctness.
- **Evidence use.** Both paradigms benefit from retrieval and tools. Multi-agent systems can divide evidence work, but shared retrieval errors may propagate across agents.
- **Reproducibility.** Single-LLM pipelines are easier to specify and reproduce. Multi-agent results depend on agent roles, communication topology, stopping rules, and debate order.
- **Scalability.** Single-LLM systems scale through larger models, better prompts, and tools. Multi-agent systems scale through additional agents, but coordination costs and convergence problems may increase.

Table 1. Comparison of single-LLM and multi-agent systems for scientific hypothesis generation.

Dimension	Single-LLM Systems	Multi-Agent Systems
Cost and efficiency	Lower computational and orchestration cost	Higher token use, latency, and coordination cost
Search breadth	Multiple samples possible through prompting	Explicit parallel exploration of diverse ideas
Criticism	Self-reflection and internal critique	Independent, role-based criticism and debate
Evidence use	Single evidence path; simpler pipeline	Evidence work can be distributed across agents
Reproducibility	Easier to reproduce	Depends on agent roles, topology, and debate configuration
Scalability	Scales through model and tool improvements	Scales through additional agents, with coordination costs
Validation	Requires external experimental validation	Also requires external experimental validation

- **Validation.** Neither architecture is sufficient without external checks. Laboratory or prospective validation is stronger than textual consensus.

Overall, the literature does not support a universal claim that multi-agent debate is superior. Multi-agent systems appear most useful when the task benefits from independent search, specialized criticism, or explicit comparison of competing hypotheses. Single-LLM systems remain attractive when cost, speed, reproducibility, and strong tool integration are more important.These differences become clearer when the systems are decomposed by function rather than treated as indivisible architectures.

3.1 A functional comparison of system components

Both paradigms may contain generation, retrieval, critique, ranking, memory, and tool use. The difference lies in how these functions are allocated. In a single-LLM pipeline, one model usually performs several functions sequentially or through structured prompting. In a multi-agent pipeline, the functions are distributed across model instances that communicate through messages or shared artifacts. Consequently, the number of agents alone is not a sufficient measure of architectural sophistication. Figure 1 visualizes this functional allocation, showing how the same broad activities can be organized within a single-LLM pipeline or distributed across specialized agents.

The first functional component is candidate generation. A single model can generate multiple candidates through temperature sampling, prompt variation, or repeated calls. This may approximate a population of agents, but the candidates are often produced under highly similar assumptions. Multi-agent systems can make independence explicit by assigning different agents different starting points, disciplinary perspectives, or evidence subsets. The benefit is greatest when the agents are encouraged to preserve competing explanations rather than immediately optimize for consensus.

The second component is evaluation. A single-LLM system can ask the model to score novelty, plausibility, and feasibility, but the scores may be internally correlated with the original generation. In a multi-agent system, separate critics can be instructed to challenge different dimensions. For example, one critic can focus on whether the claim follows from the literature, another can inspect the proposed mechanism, and a third can design a falsification test. This decomposition may

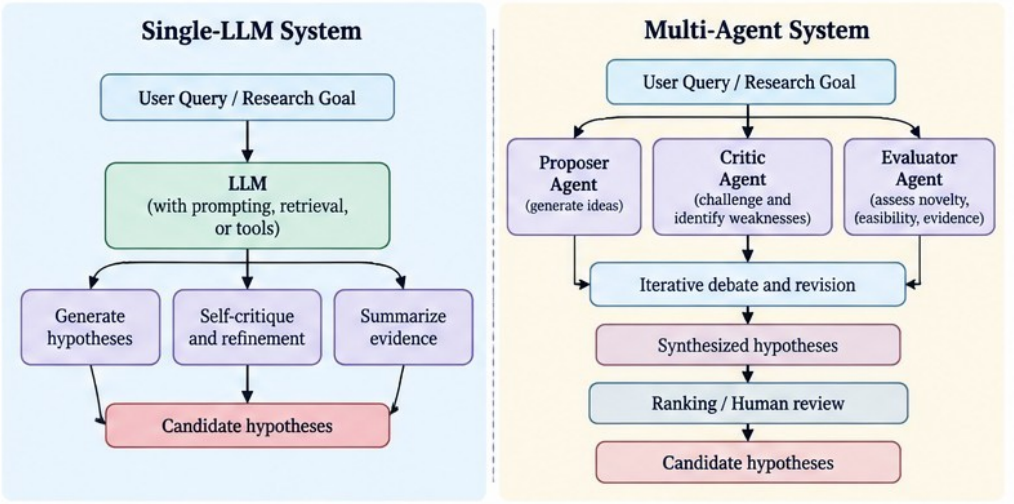


Fig. 1. Conceptual comparison of single-LLM and multi-agent systems for scientific hypothesis generation. Single-LLM systems rely on one model with prompting, retrieval or tool use, while multi-agent systems employ specialized agents that interact through debate, critique and iterative revision to produce and refine candidate hypotheses.

improve coverage, but it also creates an aggregation problem: a final controller must decide how to combine disagreement without reducing all judgments to a superficial average.

The third component is evidence management. Retrieval-augmented generation, citation tracing, structured notes, and tool calls can be used in either paradigm. A single-LLM system is often easier to audit because the evidence path is shorter. A multi-agent system can distribute retrieval and verification, but it must maintain provenance across agents. Without a shared record of which paper supports which statement, the system may produce a polished synthesis whose evidential basis is difficult to inspect.

3.2 Why more agents may help—and why they may not

The potential advantage of multiple agents can be understood partly as a reduction in the risk of shared reasoning failures. If agents independently approach a problem, disagreement may reveal an overlooked assumption or an alternative mechanism. This is particularly valuable in open-ended science, where there is rarely a single immediately verifiable answer. However, independence is a design property, not an automatic consequence of using multiple model instances. Agents sharing the same model, prompt template, retrieved papers, and evaluation rubric may reproduce the same error several times. Recent work on multi-agent debate also shows that agents can reinforce one another and converge on incorrect conclusions rather than correcting them [6].

There is also a distinction between productive disagreement and noisy disagreement. Productive disagreement is specific: an agent identifies a missing control, questions a causal interpretation, or proposes a test that could distinguish two mechanisms. Noisy disagreement merely produces different wording or unsupported objections. Studies of multi-agent research dialogues suggest that agent diversity and interaction depth can affect the quality and diversity of generated ideas, but these factors do not necessarily improve every aspect of the resulting output [21]. A useful debate protocol should therefore require each critique to identify the exact claim being challenged, the

evidence or assumption involved, and the consequence for the proposed experiment. This makes the debate more auditable and reduces the risk that verbosity is mistaken for reasoning quality.

Another limitation is premature convergence. Once one hypothesis is presented confidently or repeated by several agents, later agents may treat it as the default and modify it only superficially. Such convergence can be efficient when the leading hypothesis is correct, but dangerous when the initial proposal is flawed. Evidence of consensus and sycophantic agreement in multi-agent debate suggests that additional agents do not automatically provide independent validation [6]. Scientific systems should therefore preserve a small number of serious alternatives until evidence-based ranking or experimental testing provides a reason to eliminate them.

3.3 The role of human researchers in the comparison

The single-versus-multi-agent comparison should not be framed as a replacement contest between models and scientists. Human researchers contribute domain knowledge, tacit standards of evidence, awareness of practical constraints, and judgment about which questions matter. These contributions are difficult to capture fully in automated scores. In a single-LLM workflow, the researcher may guide the prompt, select sources, and inspect the generated hypothesis. In a multi-agent workflow, the researcher may additionally design roles, set debate rules, resolve disagreements, and approve experiments.

Human involvement is especially important at the transition from textual plausibility to scientific action. A hypothesis may be interesting but require unavailable equipment, an ethically unacceptable experiment, an unmeasured confounder, or a sample size that makes the proposed test impractical. A human-in-the-loop design can use LLMs for breadth and structured criticism while retaining human authority over interpretation, safety, and experimental prioritization. This arrangement also creates a more realistic evaluation target: the value of a system may be measured by how much it improves a scientist's decisions, not only by the quality of its standalone text.

4 Research Trends and Developments

(1) **Literature-grounded generation.** Systems increasingly retrieve papers, extract evidence, and use scientific knowledge to constrain generation [15, 22, 25].
(2) **Agent specialization.** Multi-agent systems assign roles such as generator, critic, literature analyst, experimental designer, and risk assessor [9, 19].
(3) **Closed-loop discovery.** Research systems increasingly connect hypothesis generation with coding, experiments, analysis, and revision [5, 9, 13, 23].
(4) **Benchmarking and structured evaluation.** Recent work emphasizes novelty, feasibility, validity, verifiability, and impact rather than fluency alone [10, 15, 17].
(5) **Experimental validation.** Laboratory studies are beginning to test whether LLM-generated ideas produce measurable scientific value [1, 3, 5].
(6) **Risk-aware scientific agents.** As systems become more autonomous, safety, citation faithfulness, uncertainty, and human oversight become central design requirements [2, 8, 11].

4.1 A proposed evaluation framework

A rigorous comparison should evaluate the complete hypothesis-generation process. The first dimension is novelty, measured against a time-stamped literature corpus and supplemented by expert judgment. The second is scientific validity, including whether the hypothesis is consistent with known observations and whether its causal or mechanistic explanation is coherent. The third is feasibility, which includes the availability of data, instruments, samples, and realistic experimental procedures. The fourth is predictive value: a strong hypothesis should make a sufficiently specific prediction that could be checked later.

The fifth dimension is diversity. Systems should be evaluated not only on the best idea they produce but also on the range of mechanisms and experimental directions they explore. A set of ten paraphrases should not count as ten independent discoveries. The sixth dimension is calibration: confidence should track the strength of the evidence. The seventh is efficiency, including token use, number of model calls, latency, retrieval cost, and human review time. Reporting these dimensions together would prevent a system from appearing superior merely because it uses substantially more computation.

A useful experimental design would include at least four matched conditions: a single LLM with one generation, a single LLM with multiple sampled candidates, a multi-agent system without debate, and a multi-agent system with structured debate. All conditions should receive the same model family, literature access, token budget where possible, and evaluation instructions. This design would help determine whether any observed improvement comes from agent interaction itself or simply from generating more candidates.

Evaluation should also separate retrospective and prospective settings. In retrospective evaluation, the system is given a known literature corpus and its outputs are compared with existing findings. This is useful for testing grounding and rediscovery, but it may reward ideas that are already implicit in the data. In prospective evaluation, hypotheses are generated before a later result is known and are scored against subsequent experiments or publications. Prospective evaluation is more difficult, but it better measures whether a system contributes to actual discovery.

5 Challenges and Limitations

Scientific novelty is difficult to operationalize. Similarity-based measures may reward unusual wording, whereas expert ratings can be subjective and expensive. Scientific validity often cannot be determined from the existing literature alone; it requires experiments or prospective prediction. LLMs may also hallucinate mechanisms, references, or experimental outcomes. These risks are especially serious when a multi-agent system repeats and reinforces an unsupported claim.

Multi-agent debate also incurs substantial engineering costs. Its performance depends on the number of agents, role prompts, communication structure, debate rounds, and aggregation method [18]. Consensus may suppress minority ideas, and agents may converge prematurely on a plausible but incorrect explanation. Moreover, many evaluations rely on LLM judges, which can introduce shared biases and make systems appear stronger than they are.

Finally, the available evidence is heterogeneous: studies use different domains, prompts, models, datasets, expert panels, and evaluation scales. Results from biomedical hypothesis generation therefore cannot automatically be generalized to chemistry, materials science, social science, or theoretical research. Comparisons between single- and multi-agent systems are also often confounded by unequal token budgets, retrieval access, or model capability.

5.1 Reproducibility and reporting requirements

Future studies should report sufficient implementation detail to enable meaningful comparisons. Required details include the exact model and version, prompts, temperature and sampling settings, retrieval corpus and cutoff date, number of agents, role descriptions, communication topology, number of debate rounds, stopping criteria, judge prompts, and total token expenditure. Without these details, an apparent advantage may result from hidden differences in compute or information access.

Researchers should also report negative results and failure cases, including duplicated hypotheses, unsupported citations, incorrect interpretations of papers, unresolved disagreement, and experiments that fail because the proposed mechanism was not operationalized correctly. Such reporting

is particularly important for scientific applications because a system may appear successful under language-based evaluation yet fail when its claims are tested in the laboratory.

6 Research Gaps

(7) **Controlled architecture comparisons.** Studies should compare single- and multi-agent systems with matched models, token budgets, retrieval access, and tool permissions.
(8) **Causal analysis of debate.** Experiments should isolate whether gains come from parallel sampling, role specialization, critique, additional computation, or simply more generated text.
(9) **Better diversity metrics.** Research needs measures of conceptual diversity that distinguish genuinely different mechanisms from paraphrases.
(10) **Calibration and uncertainty.** Systems should report confidence, evidence strength, disagreement, and conditions under which a hypothesis would be falsified.
(11) **Prospective and experimental validation.** Benchmarks should track whether hypotheses lead to successful future predictions or laboratory results.
(12) **Domain-sensitive evaluation.** Different fields require different definitions of novelty, feasibility, evidence, and significance.
(13) **Failure-mode reporting.** Papers should report hallucinations, unsupported citations, duplicated ideas, premature consensus, and failed experiments rather than only average scores.
(14) **Human–AI collaboration.** More work is needed on when scientists should intervene, how they should inspect agent reasoning, and how human expertise changes the quality of the final hypothesis.

6.1 Toward adaptive debate

An especially promising direction is adaptive debate rather than mandatory debate. Most candidate hypotheses may not require a large agent discussion. A lightweight single-LLM pass could first generate and filter candidates. Only proposals with high potential impact, substantial disagreement, weak evidence, or unusual safety concerns would be escalated to additional agents. This design would preserve the main efficiency advantage of single-LLM systems while allocating multi-agent computation where it is most likely to add value.

Adaptive systems could also vary the type of interaction. If the main uncertainty concerns novelty, the system should retrieve and compare prior work. If the uncertainty concerns mechanism, it should invoke a mechanistic critic. If the uncertainty concerns feasibility, it should ask an experimental-design agent to specify controls, measurements, and failure conditions. In this way, the architecture becomes evidence-driven rather than relying on a fixed number of agents or debate rounds.

7 Future Research Directions

A promising future architecture is a selective hybrid pipeline. First, several independent proposals are generated, either by one model with controlled sampling or by multiple agents. Second, each proposal is grounded in retrieved literature and linked to explicit evidence. Third, specialized critics assess novelty, validity, feasibility, safety, and experimental design. Fourth, debate is activated only for proposals with substantial disagreement or high potential value. Finally, a ranking model or human researcher selects candidates for experimental testing. This proposed workflow is summarized in Figure 2.

Future systems should preserve disagreement instead of forcing consensus. They should maintain provenance graphs linking claims to papers, observations, assumptions, and proposed tests. They

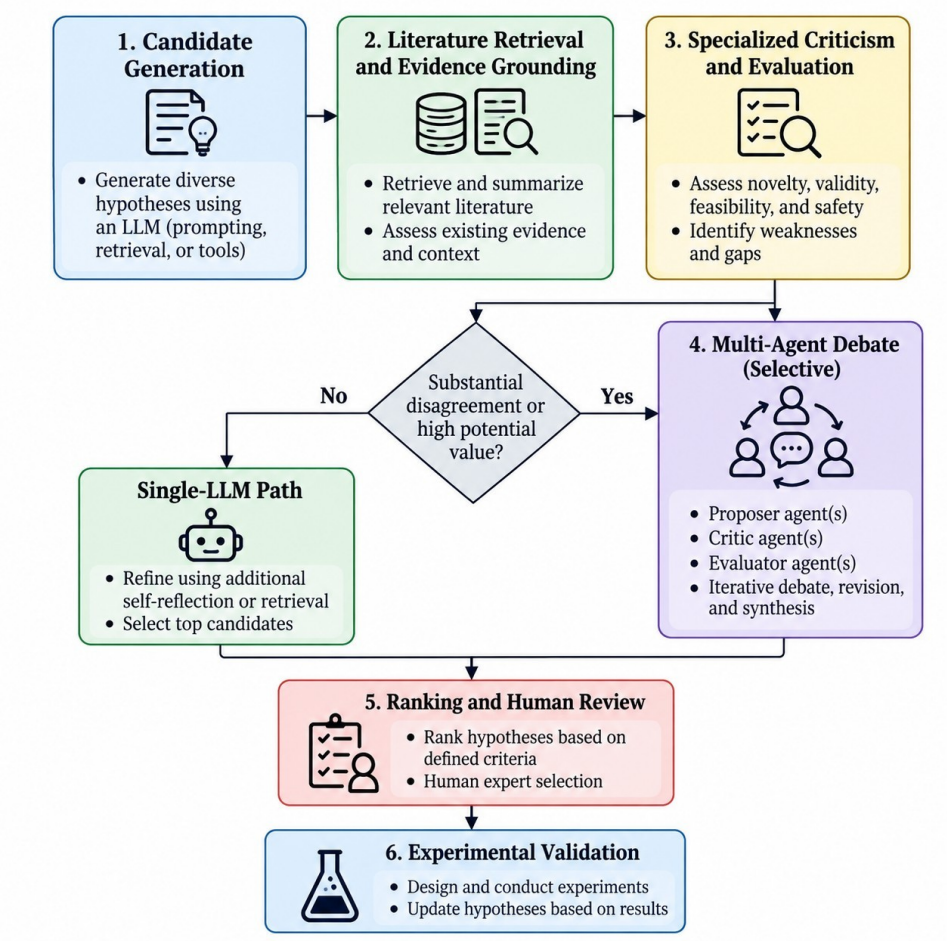


Fig. 2. Proposed hybrid workflow for scientific hypothesis generation. The system begins with candidate generation using a single LLM, followed by literature retrieval and specialized evaluation. Multi-agent debate is activated selectively when there is substantial disagreement among evaluators or high potential value in further exploration. The final candidates are ranked and reviewed by human experts before experimental validation.

should also use adaptive compute: easy cases can be handled by a single LLM, while difficult or high-impact cases receive additional agents and verification.

Evaluation should move toward prospective benchmarks. A system should generate hypotheses before a literature or experimental outcome is known, and later be scored on predictive success. This would distinguish genuine discovery from retrospective repackaging. Finally, scientific agents should be designed with risk-aware constraints, transparent uncertainty, and human approval before consequential experiments.

8 Conclusion

Single-LLM and multi-agent debate systems represent complementary strategies for scientific hypothesis generation. Single-LLM systems are efficient, reproducible, and increasingly capable

when combined with retrieval, self-reflection, and scientific tools. Multi-agent systems offer explicit parallel exploration, role specialization, and criticism, but their benefits are conditional and can be offset by cost, correlated errors, sycophancy, and premature consensus.

The literature therefore favors a conditional rather than universal conclusion. Multi-agent debate is most valuable when it introduces meaningful independence, improves evidence checking, or resolves substantive disagreement. Single-LLM systems may be preferable for rapid exploration, constrained workflows, and applications where additional interaction does not improve validation. In both cases, the decisive test is not fluent output or agent agreement, but whether a hypothesis is novel, evidence-grounded, testable, and capable of producing useful scientific knowledge.

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